

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 9

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length = $r\theta$, sector area = $\frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Find the set of values of x for which $x(x - 4) > 5$. **[3]**

2 Find the term that is independent of x in the expansion of $\left(x + \frac{2}{x^2}\right)^6$. **[4]**

3 Solve the equation $2 \cos^2 \theta = \sin \theta + 1$ for $0^\circ \leq \theta \leq 360^\circ$. **[4]**

4 A geometric progression has first term 8 and the sum of the first three terms is 14.

(a) Show that the common ratio r satisfies $4r^2 + 4r - 3 = 0$, and hence find the two possible values of r . [3]

(b) For the progression which has a sum to infinity, find this sum. [2]

5 The function f is defined by $f(x) = 2(x - 1)^2 - 5$ for $x \in \mathbb{R}$.

(a) Describe fully a sequence of three transformations which maps the graph of $y = x^2$ onto the graph of $y = f(x)$. [3]

(b) State the range of f . [1]

(c) The domain of f is now restricted to $x \geq 1$. Find an expression for $f^{-1}(x)$. [2]

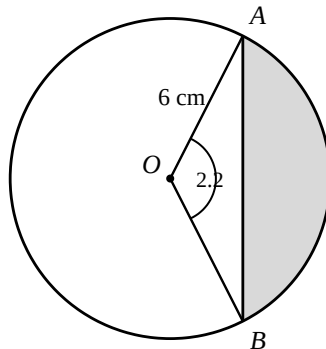
6 The points A and B have coordinates $(-3, 2)$ and $(5, 6)$ respectively.

(a) Find the coordinates of the midpoint of AB . [1]

(b) Find the equation of the line AB , giving your answer in the form $ax + by + c = 0$. [2]

(c) The point C has coordinates $(k, 0)$, where k is a constant, and angle $ACB = 90^\circ$. Find the possible values of k . [3]

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The diagram shows a circle with centre O and radius 6 cm. The points A and B lie on the circle and angle $AOB = 2.2$ radians. The minor segment bounded by the chord AB and the minor arc AB is shaded.

(a) Find the area of the **major** segment (the unshaded part of the circle). [4]

(b) Find the perimeter of the shaded minor segment. [3]

8 An open box (with no lid) has a square base of side x cm and height h cm. The volume of the box is 32 cm^3 .

(a) Show that the total external surface area, $S \text{ cm}^2$, of the box is given by $S = x^2 + \frac{128}{x}$. [2]

(b) Find the value of x for which S has a stationary value, and show that this value of S is a minimum. [4]

(c) Find the minimum value of S . [1]

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(a) Find the area of the region enclosed by the curve $y = 3x^2 - 6x$ and the x -axis. [3]

(b) The region bounded by the curve $y = \frac{2}{x}$, the x -axis and the lines $x = 1$ and $x = 2$ is rotated through 360° about the x -axis. Find the volume of the solid formed, giving your answer in terms of π . [3]

(c) Find $\int (4x + 1)^{\frac{3}{2}} dx$. [2]

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(a) Prove the identity $\tan \theta + \frac{\cos \theta}{1 + \sin \theta} \equiv \frac{1}{\cos \theta}$. [3]

(b) Hence solve the equation $\tan \theta + \frac{\cos \theta}{1 + \sin \theta} = 3 \tan \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

(c) Solve the equation $\tan\left(\frac{x}{2}\right) = -1$ for $0 \leq x \leq 2\pi$. [2]

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(a) Jasmine saves $\text{\20}$ in the first week of a savings plan. In each following week she saves $\text{\3}$ more than in the previous week.

(i) Find the amount she saves in the 12th week. [2]

(ii) Find the least number of weeks after which her total savings first exceed $\text{\$1000}$. [3]

(b) The first three terms of a geometric progression are 3, 6, 12. Find the least value of n for which the sum of the first n terms exceeds 3000. [4]

12 A circle passes through the points $O(0, 0)$, $P(6, 0)$ and $Q(0, 8)$.

(a) Find the equation of the circle. [3]

(b) Find the equation of the tangent to the circle at O . [2]

(c) Find the length of the chord of the circle which lies on the line $x = 6$. [3]

END OF PAPER (Total: 75 marks)